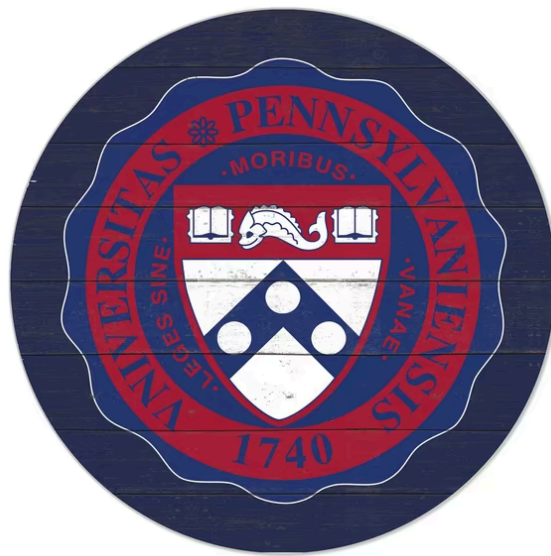


Assignment: MSSP6070_001|Assignment 1: Asking questions about Data

MSSP6070_001: Practical Programming for Data Science

PROFESSOR: Dr. Richard H Hartwell, SOCIAL POLICY AND PRACTICE

University of Pennsylvania, Philadelphia, PA, USA



Submitted By: Aatika Sinha
Graduate Student at University of Pennsylvania
E-mail: ikasinha@upenn.edu

Introduction

This case study has been based on examining academic performance among 12,411 engineering students using their academic, parental, demographic, socioeconomic, and educational characteristics. There are two stages of a student's education involved in measuring the dataset including the final year of secondary school and the final year of their professional engineering program. This project specifically applies the descriptive-statistics workflow to the “Dataset of Academic Performance evolution for Engineering Students”.

Moreover, the analysis focuses on understanding the structure and quality of the dataset, the center and spread of academic scores, score distributions and their potential outliers, distributions of important categorical variables, differences in academic performance across socioeconomic strata and parental education groups, relationships among academic score variables along with how descriptive findings can inform later data-mining work.

In addition, the analysis is descriptive and focuses on investigating whether parental education, socioeconomic groups, stratum, and gender are associated with any differences in the academic outcomes.

Key Issues - Assumptions & Limitations

A key issue of this case study is to identify differences between groups and their impact on academic outcomes. The given dataset describes one row representing one student. The academic results are obtained from two stages of education including “Saber 11” assessment and “Saber Pro” assessment with social and economic characteristics.

An important limitation concerning the dataset is the “SEL” variable. The dataset identifies “SEL” as Socioeconomic level and records values from 1 to 4, however the available data definitions do not provide additional information about how these levels were constructed or what each numerical value represents.

For this analysis, we made assumptions that the ranking is accurate and is ordered from ‘1’ representing the lowest level to ‘4’ representing the highest level. This assumption should be treated as a limitation of this analysis when interpreting the results. Also, “STRATUM” is analyzed separately because it is a categorical variable with seven levels.

Data Analysis

1. Data structure and quality

It is important to inspect data types, missing values, duplicate rows, plausible score ranges and consistency. The quality check provided the first diagnostic view of the given dataset. For the academic variables, the five 'Saber 11' subject scores are measured on a '0 to 100' scale in the source description while the 'Global Score' from the professional assessment uses a different scale. Because these variables are on different scales, they cannot be treated as additives directly. Furthermore, the source article reports **12,411 student observations and 44 variables**. It also describes the data being cleaned and encoded after linking secondary school and professional assessment records. In addition to that, descriptive checks performed are still important because the analytical dataset being used in the project should be inspected directly rather than being assumed error free.

2. Numerical Descriptive Statistics

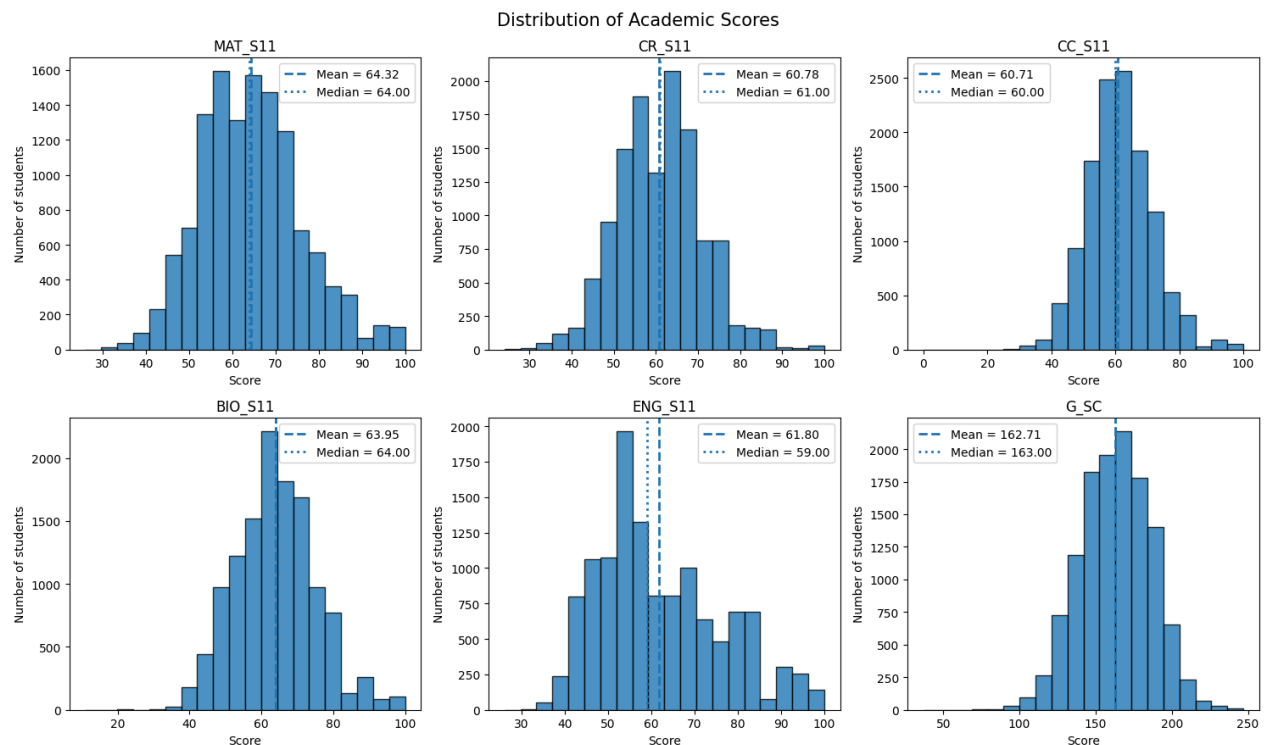
In order to describe numerical variables based on the main academic outcomes including Mathematics, Critical Reading, Citizen Competencies, Biology, English and Global Score, descriptive statistics was utilised using mean, median, mode, minimum, maximum, range, variance, standard deviation, IQR, and skewness. The mean describes the arithmetic center of a score distribution, while the median identifies the middle observation. Mean and median was compared in order to assess whether extreme observations is pulling the mean away from the center.

Moreover, the standard deviation describes the amount of variation around the mean, while the IQR describes the spread of the middle 50 percent of observations. The IQR is helpful in distributions containing unusually high and low values. The source article reported the following overall mean(s) for academic variables: Mathematics **64.32**, Critical Reading **60.78**, Citizen Competencies **60.71**, Biology **63.95**, English **61.80**, and Global Score **162.71**.

3. Visualize academic score distributions

A histogram was plotted to understand the shape of numerical distribution [Plot 1]. It identifies the concentration, skewness, gaps, and unusual observations that are difficult to see in a summary table. The mean and median was added to each distribution so that the numerical and visual descriptions can be compared. Here, the histogram shows how student scores are distributed rather than reducing each variable to one average. For the 'Saber 11' measures, the source article reported means in the **low-to-mid 60(s)**. The 'Global Score' has a higher numerical mean because it is measured on a different scale. Therefore, the visual comparison focuses on the shape and within-variable distribution, rather than interpreting the

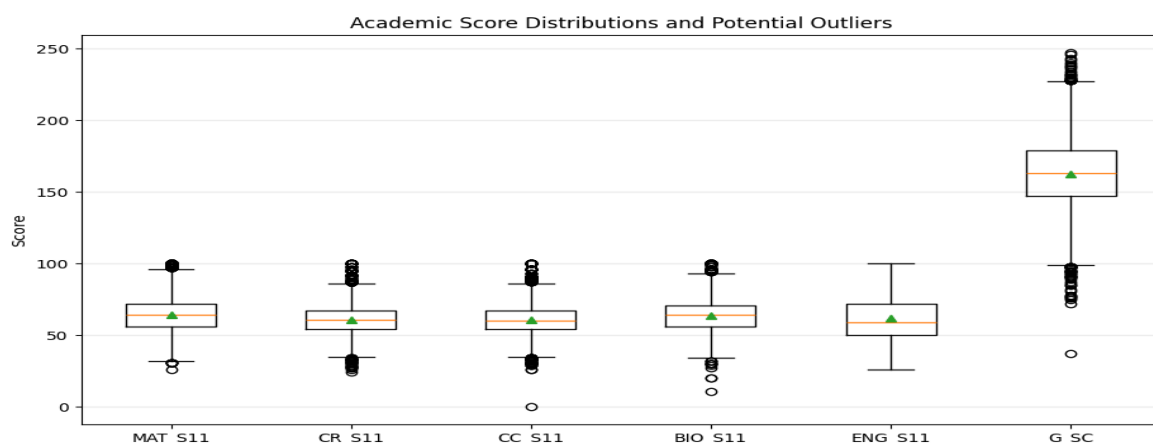
absolute heights of the score scales as directly comparable. Furthermore, it's found here that the mean and median are close, making the center of the distribution relatively similar under both measures. Also, mean and median differ more substantially, therefore histogram is inspected for skewness or extreme observations.



(Plot 1: Distribution of Academic Scores)

4. Outliers and robust statistics

Outlier is not an error automatically hence, a common '**1.5*IQR rule**' is used here to flag observations that are unusually low or high relative to each score's distribution. These observations are further investigated rather than being automatically deleted [Plot 2].



(Plot 2: Academic Score Distributions and Potential Outliers)

The boxplots and IQR table identified observations that are unusual relative to the rest of the distribution. These flagged observations should not be thought of as incorrect data, rather it is a valid student score that can be statistically unusual without representing an error in data entry. This particularly becomes important for data mining later because extreme observations can affect some statistical procedures. Therefore, here the appropriate response depends on whether the unusual values are valid observations, measurement problems or errors in data entry.

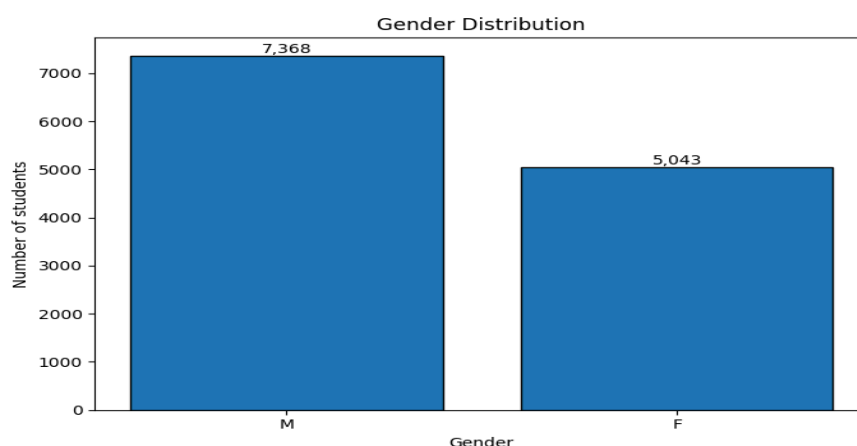
5. Categorical descriptive statistics

For categorical variables, frequency counts and percentages are more appropriate than mean. The given dataset contains several categorical characteristics that can help in describing the student population, including gender, parental education, socioeconomic stratum, school characteristics, and household resources. The source article reports that the dataset contains **12 categories for both father's education and mother's education, 7 STRATUM categories, 2 school-nature categories, and 4 school-type categories**. The education variables therefore need to be treated as categorical variables in the descriptive analysis.

This also comes with limitations on data, here two education values require particular care coded as '0' and 'Not sure'. The available codebook identifies the number of education levels but does not define '0' or provide a substantive educational rank for 'Not sure'. They are therefore retained as categories rather than being assigned an invented educational meaning.

6. Visualisation on Gender Distribution

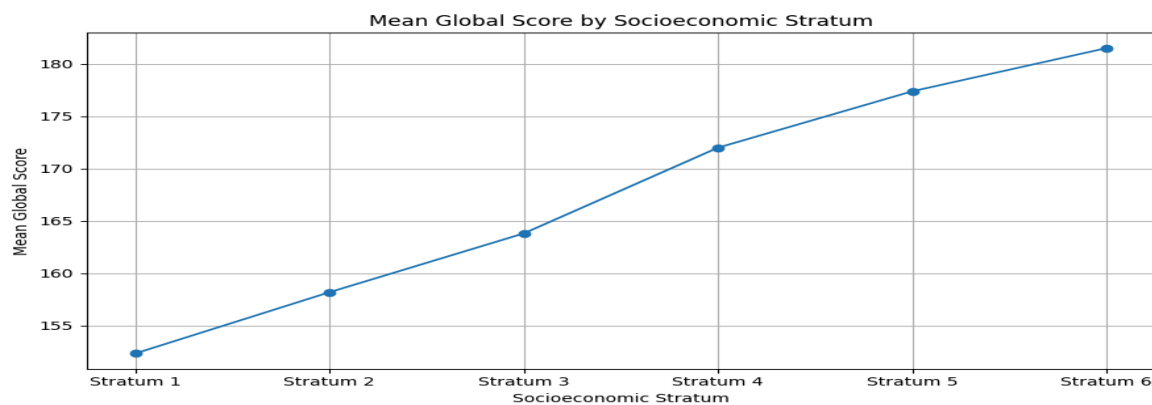
A bar chart provides a quick visual summary of a categorical variable. The source article reports 5,043 women (40.63%) and 7,368 men (59.37%) in the dataset [Plot 3].



(Plot 3: Gender Distribution)

7. Socioeconomic STRATUM and academic performance

There are substantial differences in average Global scores in the “STRATUM” categories. The ‘0’ coded STRATUM has only 14 observations while “Stratum 1” to “Stratum 6” has meaningful observation counting for our dataset. The analysis of Student Global Scores across the six strata shows a clear and consistent upward pattern [Plot 4]. The mean Global Score (G_SC) increased from 152.35 in “Stratum 1” to 158.20 in “Stratum 2”, 163.82 in “Stratum 3”, 172.00 in “Stratum 4”, 177.39 in “Stratum 5”, and 181.51 in “Stratum 6”. Thus, the difference between “Stratum 1” and “Stratum 6” is approximately 29.16 points. The median scores followed a similar pattern, increasing from 152 in “Stratum 1” to 184 in “Stratum 6”.



(Plot 4: Mean Global Score by Socioeconomic Stratum)

Moreover, the number of students was not evenly distributed across the strata. In this dataset, “Stratum 2” and “Stratum 3” have the largest number of students, with 4,029 and 4,045 observations respectively. Also, “Stratum 5” and “Stratum 6” have smaller sample sizes of 633 and 403 respectively [Table 1]. Since, the category coded as ‘0’ only contains 14 observations, it is being considered separately from the Stratum 1 to Stratum 6 progression.

STRATUM	count	mean	median	std
0	14	154	152.5	21.32171158
Stratum 1	1709	152.3528379	152	21.93740081
Stratum 2	4029	158.199305	158	22.27802876
Stratum 3	4045	163.8153276	164	21.69397037
Stratum 4	1578	172.0025349	174	21.469094
Stratum 5	633	177.3870458	179	21.5432364
Stratum 6	403	181.5111663	184	21.87940146

(Table 1: Socioeconomic STRATUM Distribution)

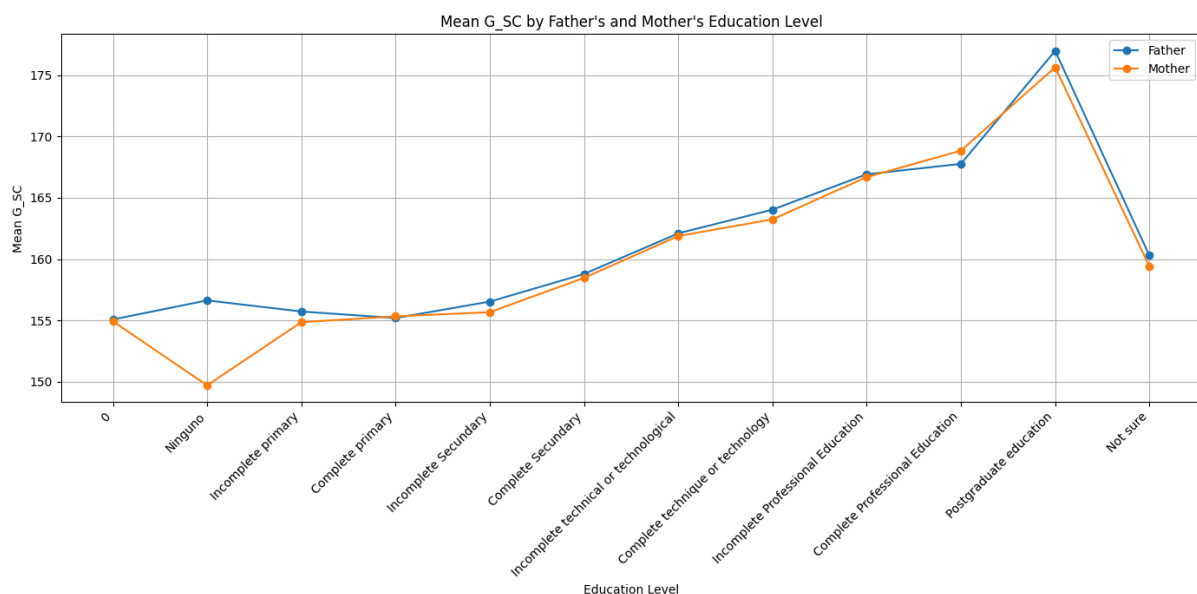
In terms of Standard Deviations, relatively similar patterns are found across “Stratum 1” and “Stratum 6”, ranging from approximately 21.47 to 22.28 points. This indicates that the variability in Global Scores (G_SC) within each stratum was broadly comparable, even though the average scores differed across stratum(s).

Overall, the descriptive results show an association between stratum and student academic performance, with higher-stratum groups having higher average Global Scores (G_SC). But still, these descriptive results alone do not establish that stratum directly causes differences in student performance.

8. Parental education influence on Student Outcomes:

The data shows a clear descriptive relationship between parental education and students ‘G_SC’ Global Scores [Plot 5]. Students whose fathers had attended the maximum level of postgraduate education had an average Global Score of approximately ‘176.99’ while the minimum education average Global score came as ‘155.08’. This represents a ‘mean’ difference of approximately ‘21.91 points’ in Father’s education considering all the ‘12’ parameters of education levels.

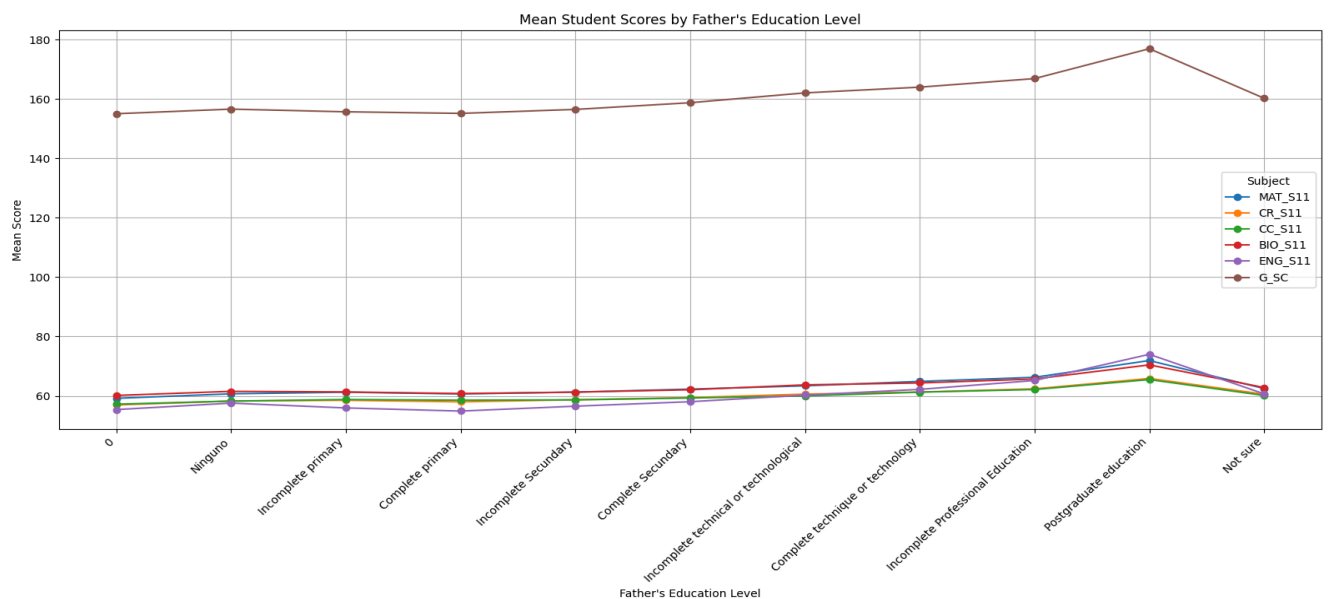
Furthermore, a similar pattern was observed for mother’s education. Students whose mothers had a maximum level of postgraduate education had an average Global Score of approximately ‘175.64’, while the minimum education average Global score came as ‘149.70’. This represents a difference of approximately ‘25.93 points’ in mother’s education considering all the ‘12’ parameters of education levels.



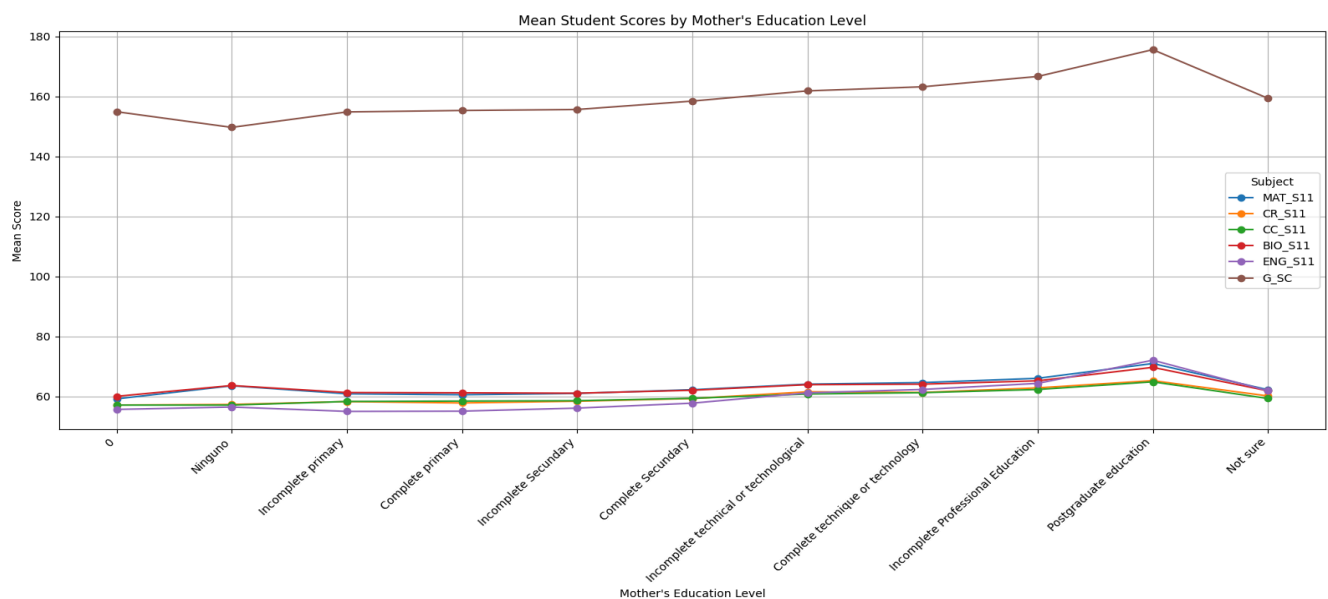
(Plot 5: Mean G_SC by Father’s and Mother’s Education Level)

Moreover, to examine whether parental education is associated with differences in student academic performance, student’s scores were compared across the different education categories of both fathers and mothers[Plot 6,7]. The analysis considered

the student's overall Global Score (G_SC) as well as individual academic subject scores, including Mathematics (MAT_S11), Critical Reading (CR_S11), Critical Competency (CC_S11), Biology (BIO_S11), and English (ENG_S11).



(Plot 6: Mean Student Scores by Father's Education Level)



(Plot 7: Mean Student Scores by Mother's Education Level)

Overall, the results suggest that a student's academic performance differs significantly across parental education categories. The comparison of individual subject scores can further show whether these differences are consistent across mathematics, Critical Reading, Critical Competency, Biology, and English. A rank-based analysis also showed a positive association between education level and Global score for both father(s) and mother(s). However, these results indicate an

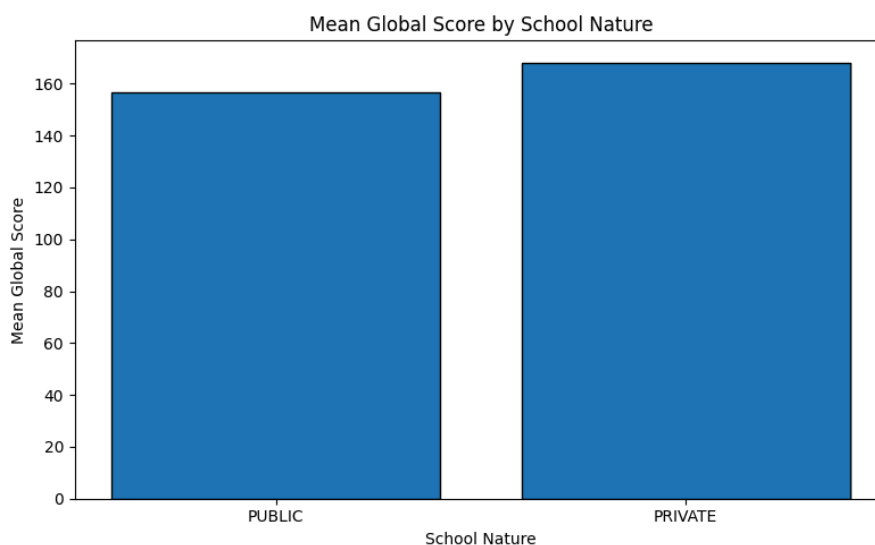
association between parental education and student outcomes and do not establish that parental education directly causes differences in academic performance since other socioeconomic (SEL), educational, and demographic factors may also contribute.

9. One-way ANOVA

Additionally, to assess whether student outcomes differed across parental-education categories, separate one-way ANOVAs were conducted for father's and mother's education. For father's education, the ANOVA produced an F-statistic of 93.01 with a p-value significantly less than 0.05. For mother's education, the F-statistic was 92.31 with a p-value significantly less than 0.05 as well. Both p-values are substantially below the 0.05 significance level, indicating statistically significant differences in mean Global scores (G_SC) across the parental-education categories for both father(s) and mother(s). The ANOVA results established that at least some education groups differ in their average student outcomes, but they do not identify which specific groups differ or establish a causal relationship between parental education and student performance.

10. Additional group comparisons

The same descriptive approach can be used to examine other categorical characteristics in the dataset. These comparisons are useful for identifying patterns that may be helpful for further investigation. For example, comparing mean Global Score by school nature [Plot 8]. These visualisations identified difference in average Global Scores across gender, school characteristics, and selected household-resource categories.

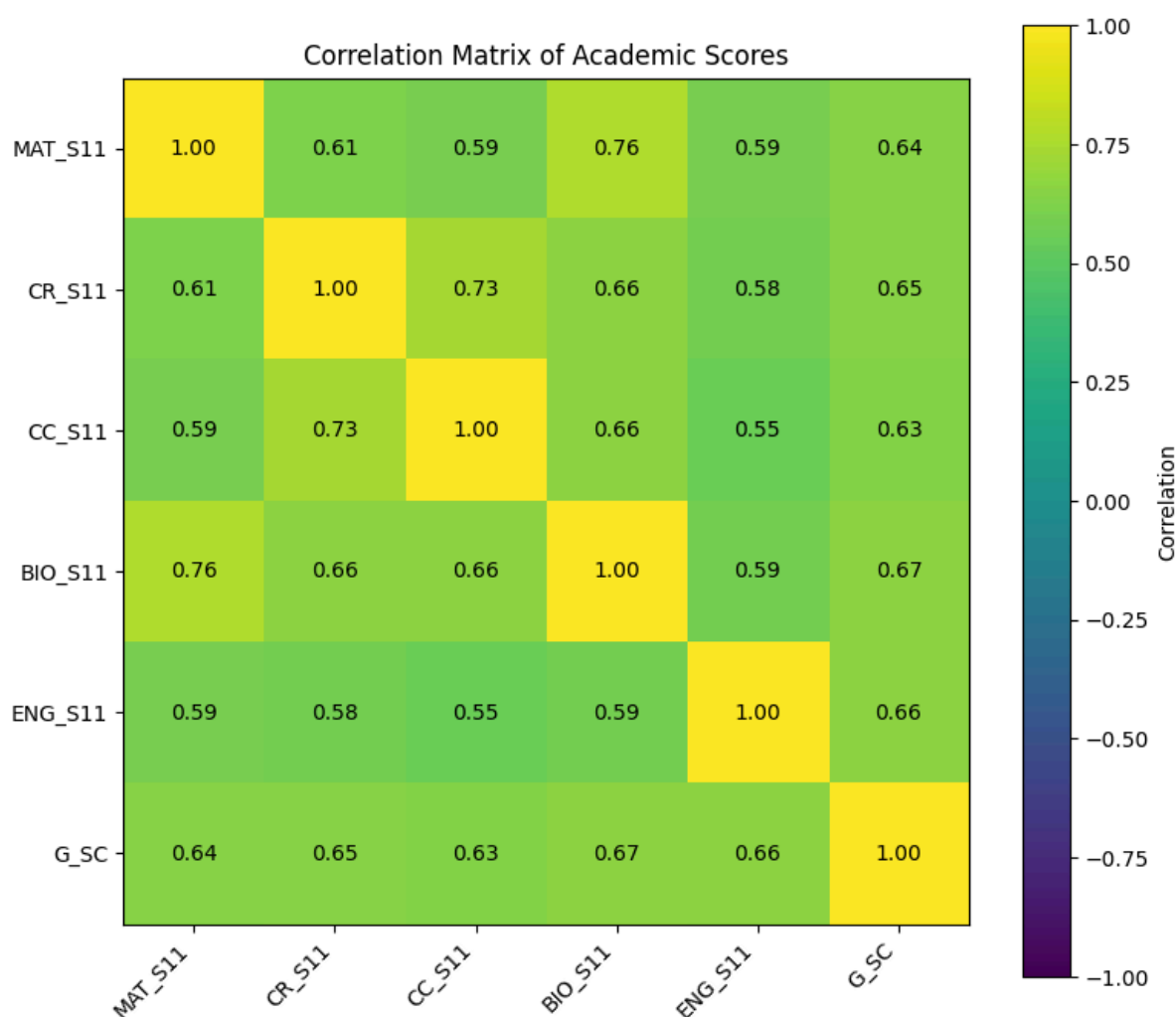


(Plot 8: Mean Global Score by School Nature)

11. Relationships among academic scores

Correlation summarises the direction and strength of a linear relationship between two numerical variables. Here, making use of the correlation matrix to allow several score relationships to be examined at once[Plot 9].

Because the academic measures come from related standardized assessments, strong correlations indicate that students who score relatively higher on one measure also tend to score relatively higher on another measure. However, correlation alone doesn't establish why the relationship exists.



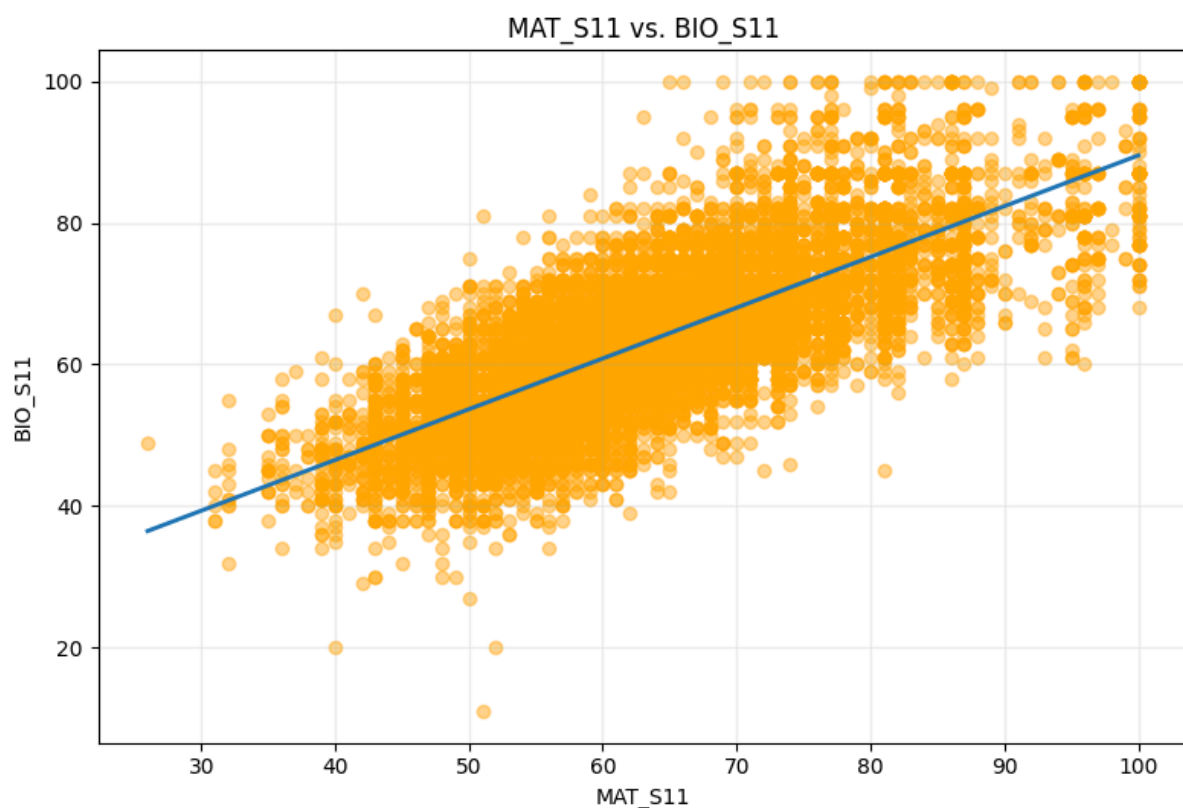
(Plot 9: Correlation Matrix of Academic Scores)

The correlation matrix here shows which academic measures move together linearly. The relationship is interpreted in terms of direction and strength rather than as causal effects. Furthermore, the source dataset describes the academic variables as measures of related competencies assessed through national standardised examinations. Since the variables measure different competencies, correlation can provide useful evidence about whether students' relative performance tends to be

consistent across subjects. In addition to that, a strong correlation does not mean that one subject score causes another subject score. It reflects shared underlying academic skills, common assessment conditions, or other characteristics of the students.

12. Scatterplot for investigating correlation

The scatterplot provides a visual check of the strongest observed linear association. It allows us to see whether the relationship is approximately linear, whether there are clusters, and whether unusual observations appear to drive the correlation. Here, the strongest linear association was found between Mathematics (MAT_S11) and Biology (BIO_S11) scores[Plot 10]. This illustrates that numerical summaries and graphics should be used together. A correlation coefficient compresses a relationship into one number, while a scatterplot restores the information about the underlying pattern.



Variables: MAT_S11 and BIO_S11

Correlation: 0.763

(Plot 10: MAT_S11 and BIO_S11 Scatterplot)

Findings & interpretations

Combining the descriptive-statistics workflow to the engineering-student dataset, following has been discovered:

1. **Dataset and academic performance:** The dataset contains 12,411 student records and 44 variables, combining academic, social and economic information. The academic measures includes five 'Saber 11' subject scores and later professional-assessment measures.
2. **Center and spread:** The descriptive statistics show the center and variability of Mathematics, Critical Competencies, Biology, English, and the Global Score. Mean, median, standard deviation, IQR, and skewness provide complementary information about student performance.
3. **Distributions and Outliers:** Histograms and boxplots show the shape and unusual observations in the score distributions. The $1.5 \times \text{IQR}$ rule is used only to flag potential outliers. Moreover, flagged observations are not automatically treated as errors.
4. **Socioeconomic stratum:** Mean Global Scores increase across the ordered 'Stratum 1' to 'Stratum 6' groups in the working analysis, from approximately 152.35 to 181.51. The results therefore show a substantial descriptive difference across socioeconomic strata, while the category coded '0' is treated separately because it contains very few observations.
5. **Parental education:** Mean Global Scores differs across parental education categories for both fathers and mothers. In the working analysis, the father-education comparison ranges from approximately 155.08 to 176.99, while the mother-education comparison ranges from approximately 149.71 to 175.64. The one-way ANOVA also indicated statistically significant differences across the parental-education groups. These results support an association between parental education categories and student performance, but they do not establish causation.
6. **Relationship among scores:** The correlation matrix and scatterplot provide a second perspective on academic performance by showing how subject scores move together. These relationships can inform later modeling, feature selection, and interpretation.

Conclusion

Averages alone are not sufficient. Efficient data analysis combines numerical summaries, visualisations, category distributions, group comparisons, and contextual judgement. Descriptive analysis provides the foundation for later data mining because it helps in identifying missingness, unusual values, distributional patterns, category imbalance, group differences, and relationships among variables before a predictive or classification model is built.

Discussion of association

1. Which score variables have the largest difference between their mean and median? What might the distribution suggest about this difference?
2. Why should '0' and 'Not sure' in parental education be retained as categories rather than assigned an unsupported educational rank?
3. Does the scatterplot support the pattern suggested by the correlation coefficient?
4. Which potential outliers should be investigated before deleting any observations?
5. What additional variables should be included before making a casual claim about parental education, socioeconomic stratum, or school characteristics and academic performance?

References

Delahoz-Dominguez, E., Zuluaga, R., & Fontalvo-Herrera, T. (2020). *Dataset of academic performance evolution for engineering students*. Data in Brief, 30, 105537. <https://doi.org/10.1016/j.dib.2020.105537>

The dataset contains 12,411 observations and 44 variables and combines academic, social and economic information for engineering students in Columbia.

Appendix A - Jupyter Notebook

- Python code - The complete Python/Jupyter Notebook containing the code, calculations, statistical analyses, and visualisations used in this Case Study is submitted as a separate file along with this written report.
- File: MSSP6070_001DescriptiveStatsEngg_PrfrmAssignment1_Aatika.ipynb